I- Partial fractions

Investigation

Show that $\frac{7x+1}{x^2+x-6} = \frac{3}{x-2} + \frac{4}{x+3}$

a) Methods

Property

Step 1 : Factorise the denominator.

Step 2 : Write a sum of partial fractions with the factors as denominators and unknown numerators.

Step 3 : Multiply both sides by the denominator.

Step 4 : Find the unknown numerators by substituting the zeros of the denominators.

Example

Write as a sum of partial fractions :

$$\frac{7x+1}{x^2+x-6}.$$

Example

Write

$$\frac{33-12x}{(x+1)(x-2)^2} \quad \text{as the sum of partial fractions in the form} \quad \frac{A}{x+1}+\frac{B}{x-2}+\frac{C}{(x-2)^2}.$$

b) Exercises Set I

EXERCISE 1 Show that :

1. $\frac{3}{x^2+5x+4}$ can be written as the sum of partial fractions $\frac{-1}{x+4}+\frac{1}{x+1}$
2. $\frac{9-x}{x^2+3x-10}$ can be written as the sum of partial fractions $\frac{-2}{x+5}+\frac{1}{x-2}$
3. $\frac{3x-23}{2x^2-5x-12}$ can be written as the sum of partial fractions $\frac{5}{2x+3}-\frac{1}{x-4}$

EXERCISE 2 Write as the sum of partial fractions :

1. $\frac{1}{x^2+3x+2}$
2. $\frac{x}{x^2-5x+6}$
3. $\frac{3x}{x^2-2x-8}$
4. $\frac{3x+2}{x^2-4}$
5. $\frac{2x-1}{x^2-2x-3}$
6. $\frac{2-5x}{x^2+4x-12}$
7. $\frac{7x+10}{2x^2+5x+3}$
8. $\frac{x+9}{3x^2-5x-2}$
9. $\frac{2x+27}{8x^2+18x-5}$

EXERCISE 3 Show that

$$\frac{5x^2+11x+16}{(x-5)(x+2)^2} \quad \text{can be written as the sum of partial fractions} \quad \frac{4}{x-5}+\frac{1}{x+2}-\frac{2}{(x+2)^2}.$$

EXERCISE 4 Write

$$\frac{-3x-4}{(x+2)(x+3)^2} \quad \text{as the sum of partial fractions in the form} \quad \frac{A}{x+2}+\frac{B}{x+3}+\frac{C}{(x+3)^2}.$$

EXERCISE 5

1. Show that $(x+3)(x-1)^2 = x^3+x^2-5x+3$
2. Write $\frac{19x+9}{x^3+x^2-5x+3}$ as the sum of partial fractions.

EXERCISE 6

1. Show that $(2x-1)(x^2+x+4) = 2x^3+x^2+7x-4$
2. Write
$$\frac{10x^2-3x+18}{2x^3+x^2+7x-4} \quad \text{in the form} \quad \frac{A}{2x-1}+\frac{Bx+C}{x^2+x+4}.$$

II- Rational functions

a) Asymptotes

Investigation

For a function of the form
$$y = \frac{ax+b}{cx^2+dx+e}, \quad c \neq 0 :$$

1. What is the horizontal asymptote ?
2. How can we determine the vertical asymptotes ?

Property

For a function of the form

$$y = \frac{ax + b}{cx^2 + dx + e}, \quad c \neq 0 :$$

— The horizontal asymptote is $y = 0$

— The vertical asymptotes are the zeros of the denominator $cx^2 + dx + e$

Example

Consider the function

$$y = \frac{2x - 6}{x^2 - 3x - 4}.$$

— Find the asymptotes of the function.

— Find the axes intercepts.

— Draw a sign diagram of the function.

— Hence discuss the behaviour of the function near the asymptotes.

— Sketch the function, showing the features you have found.

Investigation

Consider the function

$$y = \frac{6x^2 + x - 3}{2x + 1}.$$

1. State the equation of the vertical asymptote.
2. Show that
$$\frac{6x^2 + x - 3}{2x + 1} = 3x - 1 - \frac{2}{2x + 1}.$$
3. Use technology to sketch
$$y = \frac{6x^2 + x - 3}{2x + 1} \quad \text{and} \quad y = 3x - 1$$
on the same set of axes.
4. Describe the behaviour of
$$y = \frac{6x^2 + x - 3}{2x + 1}$$
as $|x| \to \infty$.

Property

To help us describe graphs of the form

$$y = \frac{ax^2 + bx + c}{dx + e},$$

we use polynomial division to write them in the form

$$y = px + q + \frac{r}{dx + e}.$$

— The function is undefined when $dx + e = 0$, so the graph has the vertical asymptote

$$x = -\frac{e}{d}.$$

— As $|x| \to \infty$, the term

$$\frac{r}{dx + e} \to 0,$$

and so the graph gets closer to the line $y = px + q$. The graph has the oblique asymptote $y = px + q$.

Remark. An oblique asymptote is an asymptote which is neither vertical nor horizontal.

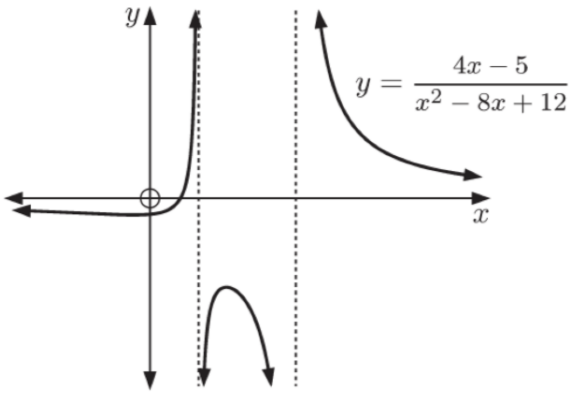
b) Exercises Set II

EXERCISE 7

The graph of $y = \dfrac{4x - 5}{x^2 - 8x + 12}$ is shown alongside.

a Find the asymptotes of the function.

b Find the axes intercepts.



EXERCISE 8

For each of the following functions :

i. Find the equations of the asymptotes.

ii. Find the axes intercepts.

iii. Draw a sign diagram of the function.

iv. Hence discuss the behaviour of the function near its asymptotes.

v. Sketch the graph of the function.

a $y = \frac{x-1}{(x+3)(x-2)}$ **b** $y = \frac{2x-8}{(x-1)(x-3)}$ **c** $f(x) = \frac{5-3x}{(x+2)(x-4)}$ **d** $f(x) = \frac{-x-4}{x^2+3x}$

e $y = \frac{2x+5}{x^2-4x-5}$ **f** $y = \frac{12-4x}{x^2-3x-18}$ **g** $f(x) = \frac{5x-9}{2x^2+10x-12}$ **h** $y = \frac{3x+6}{x^2+2x+1}$

i $f(x) = \frac{4-x}{2x^2-9x-5}$ **j** $y = \frac{6x+15}{3x^2+2x-8}$ **k** $f(x) = \frac{7-4x}{21+8x-4x^2}$ **l** $y = \frac{1-3x}{9x^2-12x+4}$

EXERCISE 9

Consider the function $y = \frac{4x-6}{x^2-5x+7}$

1. State the equation of the horizontal asymptote.
2. Explain why the function has no vertical asymptotes.
3. Find the axes intercepts.
4. Use technology to help sketch the function.

EXERCISE 10

The function $f(x) = \frac{8x-3}{cx^2+dx+9}$ has one vertical asymptote, and passes through the point $(1, 5)$.

1. Show that $f(x)$ can take two possible forms. Find the vertical asymptote of $f(x)$ in each case.
2. Sketch the possible graphs of $y = f(x)$ on the same set of axes.

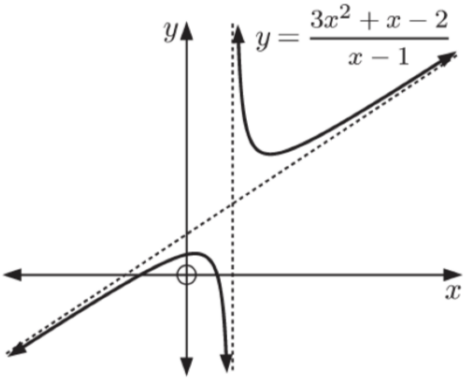
EXERCISE 11

The graph of $y = \frac{3x^2+x-2}{x-1}$ is shown alongside.

a Find the equation of the vertical asymptote.

b Find the axes intercepts.

c Write the function in the form $y = px + q + \frac{r}{x-1}$, and hence find the equation of the oblique asymptote.



EXERCISE 12

For each of the following functions:

i Find the equation of the vertical asymptote.
ii Find the axes intercepts.
iii Find the oblique asymptote.
iv Draw a sign diagram of the function.
v Hence discuss the behaviour of the function near its asymptotes.
vi Sketch the graph of the function.

a $y = \dfrac{x^2 + x - 6}{x - 1}$ b $f(x) = \dfrac{-x^2 + 3x - 2}{x}$ c $y = \dfrac{2x^2 - 8x + 8}{x - 3}$

d $f(x) = \dfrac{-6x^2 + 5x}{2x + 1}$ e $y = \dfrac{-6x^2 - 4x - 1}{3x + 2}$ f $f(x) = \dfrac{8x^2 - 19x - 15}{1 - 2x}$

III- Even and odd functions

a) Definitions and examples

Definition

A function $f(x)$ is even if $f(-x) = f(x)$ for all x in the domain of f.

The graph of an even function is symmetrical about the y-axis.

Definition

A function $f(x)$ is odd if $f(-x) = -f(x)$ for all x in the domain of f.

The graph of an odd function has rotational symmetry about the origin.

Example

Determine whether each function is even, odd, or neither :

1. $f(x) = \dfrac{1}{x}$
2. $g(x) = x^4 + x^2$
3. $h(x) = x^2 - 2x + 3$

Example

Prove that the sum of two odd functions is also an odd function.

b) Exercises Set III

EXERCISE 13

Determine whether each function is even, odd, or neither:

a $f(x) = 5x$ b $f(x) = -4x + 3$ c $f(x) = \dfrac{3}{x^2 - 4}$

d $f(x) = 2x^3 - \dfrac{5}{x}$ e $f(x) = x^2 + \dfrac{7}{x^2} - 3$ f $f(x) = \sqrt{x}$

EXERCISE 14

Suppose $g(x)$ is an odd function, and $(4, 6)$ and $(-1, 2)$ lie on the graph of $y = g(x)$. Write down two other points which lie on the graph of $y = g(x)$.

EXERCISE 15

Suppose $f(x) = (2x + 3)(x + a)$, $a \in \mathbb{R}$ is an even function. Find the value of a.

EXERCISE 16

a If the quadratic function $f(x) = ax^2 + bx + c$, $a \neq 0$ is an even function, prove that $b = 0$.

b Under what conditions is the cubic function $g(x) = ax^3 + bx^2 + cx + d$, $a \neq 0$, an odd function?

c Under what conditions is the quartic function $h(x) = ax^4 + bx^3 + cx^2 + dx + e$, $a \neq 0$, an even function?

EXERCISE 17

Decide whether each of the following is even, odd, or neither. Explain how this is observed in the graph of the function.

a $f(x) = \cos x$ b $f(x) = \sin x$ c $f(x) = \tan x$

EXERCISE 18

For what values of k is $f(x) = \cos(x - k)$:

a even b odd c neither?

EXERCISE 19

Prove that:

a the sum of two even functions is an even function

b the difference between two odd functions is an odd function

c the product of two odd functions is an even function.

EXERCISE 20

Suppose $f(x)$ is an even function and $g(x)$ is an odd function. Determine whether the function $(f \circ g)(x)$ is even, odd, or neither.

IV- Squared functions

a) Intercepts, Asymptotes, and Intersections

Investigation

1. Sketch each pair of functions on the same set of axes. Include all axes intercepts and asymptotes.

   (a) $y = x + 1$ and $y = (x + 1)^2$

   (b) $y = x^2 - 4$ and $y = (x^2 - 4)^2$

   (c) $y = \dfrac{2}{x}$ and $y = \left(\dfrac{2}{x}\right)^2$

   (d) $y = \dfrac{1}{x - 3}$ and $y = \dfrac{1}{(x - 3)^2}$

   (e) $y = 2^x - 1$ and $y = (2^x - 1)^2$

   (f) $y = \log x$ and $y = (\log x)^2$

2. Determine the points where each pair of functions intersect. What do you notice ?

3. What other observations can you make about the graph of $y = [f(x)]^2$?

Property

When $y = [f(x)]^2$ is graphed from $y = f(x)$:

— The points with y-coordinate 0 or 1 are invariant.

— The graph of $y = [f(x)]^2$ touches the x-axis at its x-intercepts.

— The graph of $y = [f(x)]^2$ lies above or on the x-axis for all x.

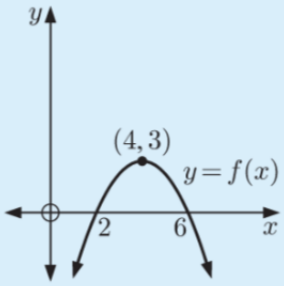
— The vertical asymptotes of $y = f(x)$ are also vertical asymptotes of $y = [f(x)]^2$.

— When $-1 < f(x) < 1$, $y = [f(x)]^2$ is closer to the x-axis than $y = f(x)$.

— When $|f(x)| > 1$, $y = [f(x)]^2$ is further from the x-axis than $y = f(x)$.

Example

For the graph of $y = f(x)$ alongside, draw the graph of $y = [f(x)]^2$.



$y = f(x)$ cuts the x-axis at 2 and 6, so $y = [f(x)]^2$ touches the x-axis at 2 and 6.

$y = f(x)$ and $y = [f(x)]^2$ also intersect when $y = 1$.

$y = f(x)$ has a turning point at $(4, 3)$, so $y = [f(x)]^2$ has a turning point at $(4, 9)$.

